

the most ancient algorithms have their loopholes, any generalization here is basically a waste on the energy that was spent making that allegation. There are some rather credible schools of thought that their 'weak' algorithms were made deliberately, so as to 'pass through all of the international export laws and establish a global base for a new protocol – 802.11'. Nope, even we don't know how that makes sense. Never mind. So the legend says that if one were to attempt breaking such an encryption by the use of brute force technique (because no other technique holds any proof of its optimality), it would turn out to be a very, very difficult task. Simply fulfilling the memory requirements of such an attack would need a sizable number of terabytes of storage, coupled with a few thousand nodes to supply the resources. Incidentally, the record for the largest RC5 (Rivet Cipher 5) code broken is 64-bits, and that was about 10 years ago! A 128-bit code is much, much stronger than a 64-bit code that we just talked about – so much so the difficulty in cracking them is of the order of a few billion times over the 64-bit code.

Obviously, over time (as in, a few hundred years), the capacities of cracking systems will improve, and they might be able to shave off a number or two from those requirements, but by that time, it would probably not matter that you bought that plasma screen TV online, last diwali. So you can do all of that safe in the knowledge that by the time your information is breached, it would have been long useless. And here we're only talking 128-bit cracking!

There's data out there that would still need to be safeguarded after a few hundred years (government data, or corporate records spring to mind). For those, there's always 256-bit encryptions. Just to give you a perspective of the kind of power that would need to be cracked open, 2^{256} is a lot more than the number of atoms – in the Universe!

So we now know two things here:

First, that encryption techniques are pretty awesome, and would take some beating. Some beating, indeed.

Secondly, and very importantly (for those who are/ hope to be in a technical field) -

"Hashing is not encryption".

Words to live by.

Now let us shake some ground that we've laid out so confidently for you – encryption can, theoretically, be broken. And very easily, at that. All it would take is an implementation of something known as a quasi meltoromotoratron. Okay, not that, but just as sci-fi: quantum computer.